Algebra Practice Worksheet

Topics: Equations · Absolute Value · Inequalities

Name: _____ Date: _____ Period: _____

Instructions: Show all steps clearly in the space provided. Write your final answer on the answer line. Check your answer when possible.

Golden Rule: Whatever you do to one side of an equation or inequality, do the **same** to the other side.

Inequality Warning: When you multiply or divide both sides by a **negative** number, **flip the inequality sign!** Example: $-2x > 6 \Rightarrow x < -3$

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	Total	**62 problems**

Topic 1 Solving Equations

Goal: Get x alone on one side by undoing what is done to it.

Form	**Undo with...**	**Quick Example**
$x + a = b$	subtract a from both sides	$x + 4 = 9 \Rightarrow x = 5$
$x - a = b$	add a to both sides	$x - 3 = 8 \Rightarrow x = 11$
$ax = b$	divide both sides by a	$4x = 20 \Rightarrow x = 5$
$\frac{x}{a} = b$	multiply both sides by a	$\frac{x}{3} = 5 \Rightarrow x = 15$

Part A One-Step Equations (10 problems)

Example: Solve $x + 6 = 10$

The number 6 is added to x. Undo it by subtracting 6 from both sides:

$$x + 6 \; - \; 6 \; = \; 10 \; - \; 6 \qquad \Longrightarrow \qquad \boxed{x = 4}$$

Check: $4 + 6 = 10$ ✓

1. $x + 5 = 9$

Undo: _____

$x =$ _____

Check: _____

2. $x - 3 = 7$

Undo: _____

$x =$ _____

Check: _____

3. $2x = 14$

Undo: _____

$x =$ _____

Check: _____

4. $\frac{x}{4} = 3$

Undo: _____

$x =$ _____

Check: _____

5. $x + 8 = 2$

Undo: _____

$x =$ _____

Check: _____

6. $x - 6 = -1$

Undo: _____

$x =$ _____

Check: _____

7. $3x = -12$

Undo: _____

$x =$ _____

Check: _____

8. $\frac{x}{5} = -2$

Undo: _____

$x =$ _____

Check: _____

9. $-x = 9$ *(Hint: divide by -1)*

$x =$ _____

Check: _____

10. $x + 4 = -3$

Undo: _____

$x =$ _____

Check: _____

Part B Two-Step Equations (10 problems)

Example: Solve $3x - 5 = 7$

Step ① Add 5 to both sides (undo the subtraction):

$$3x - 5 + 5 = 7 + 5 \qquad \Longrightarrow \qquad 3x = 12$$

Step ② Divide both sides by 3 (undo the multiplication):

$$\frac{3x}{3} = \frac{12}{3} \qquad \Longrightarrow \qquad \boxed{x = 4}$$

Check: $3(4) - 5 = 12 - 5 = 7$ ✓

1. $2x + 3 = 11$

Step ①: $2x =$ _____

Step ②: $x =$ _____

Check: _____

2. $3x - 5 = 10$

Step ①: $3x =$ _____

Step ②: $x =$ _____

Check: _____

3. $4x + 1 = 17$

Step ①: $4x =$ _____

Step ②: $x =$ _____

Check: _____

4. $2x - 7 = 1$

Step ①: $2x =$ _____

Step ②: $x =$ _____

Check: _____

5. $5x + 2 = -8$

Step ①: $5x =$ _____

Step ②: $x =$ _____

Check: _____

6. $3x - 4 = -13$

Step ①: $3x =$ _____

Step ②: $x =$ _____

Check: _____

7. $\dfrac{x}{2} + 3 = 7$

Step ①: $\frac{x}{2} =$ _____

Step ②: $x =$ _____

Check: _____

9. $2x + 9 = 3$

Step ①: $2x =$ _____

Step ②: $x =$ _____

Check: _____

8. $\frac{x}{3} - 2 = 4$

Step ①: $\frac{x}{3} =$ _____

Step ②: $x =$ _____

Check: _____

10. $4x - 3 = 9$

Step ①: $4x =$ _____

Step ②: $x =$ _____

Check: _____

Topic 2 Solving Absolute Value Equations

Key Fact: $|x|$ = distance from x to 0. Distance is never negative.

$c > 0$: $|E| = c \implies E = c$ or $E = -c$ (two solutions)

$c = 0$: $|E| = 0 \implies E = 0$ (one solution)

$c < 0$: $|E| = c \implies$ **No solution**

3 Steps: ① Isolate $|\cdot|$ ② Split: $E = c$ or $E = -c$ ③ Solve each equation

Type 1 Example (Basic): Solve $|x| = 4$

Already isolated. Split: $x = 4$ or $x = -4$

Think of it on a number line: 4 units left of 0 and 4 units right of 0.

$$\boxed{x = 4 \quad \text{or} \quad x = -4}$$

Type 2 Example (Shift): Solve $|x - 3| = 5$

Already isolated. Split:

$x - 3 = 5 \Rightarrow x = 8$ or $x - 3 = -5 \Rightarrow x = -2$

$$\boxed{x = 8 \quad \text{or} \quad x = -2}$$

Check: $|8 - 3| = |5| = 5$ ✓ $|-2 - 3| = |-5| = 5$ ✓

Type 3 Example (Isolate First): Solve $|x + 4| + 2 = 9$

Step ① Subtract 2 from both sides: $|x + 4| = 7$

Step ② Split: $x + 4 = 7 \Rightarrow x = 3$ or $x + 4 = -7 \Rightarrow x = -11$

$$\boxed{x = 3 \quad \text{or} \quad x = -11}$$

Type 4 Example (Challenge): Solve $|2x - 1| = 7$

Case 1: $2x - 1 = 7 \Rightarrow 2x = 8 \Rightarrow x = 4$

Case 2: $2x - 1 = -7 \Rightarrow 2x = -6 \Rightarrow x = -3$

$$\boxed{x = 4 \quad \text{or} \quad x = -3}$$

Problems 1–3: Type 1 Problems 4–6: Type 2 Problems 7–8: Type 3 Problems 9–10: Type 4

1. $|x| = 5$
Split: $x =$ _____ or $x =$ _____
Answer: _____

2. $|x| = 0$
(Hint: $c = 0$, one answer)
Answer: _____

3. $|x| = -2$
(Hint: can distance be negative?)
Answer: _____

4. $|x - 3| = 4$
$x - 3 =$ _____ or $x - 3 =$ _____
$x =$ _____ $x =$ _____
Answer: _____

5. $|x + 2| = 6$
$x + 2 =$ _____ or $x + 2 =$ _____
$x =$ _____ $x =$ _____
Answer: _____

6. $|x - 1| = 0$
(One solution)
$x - 1 =$ _____ $\Rightarrow$ $x =$ _____
Answer: _____

7. $|x + 4| + 2 = 9$
Step ①: $|x + 4| =$ _____
$x + 4 =$ _____ or $x + 4 =$ _____
$x =$ _____ $x =$ _____
Answer: _____

8. $2|x| = 10$
Step ①: $|x| =$ _____
Split: $x =$ _____ or $x =$ _____
Answer: _____

9. $|2x - 3| = 7$
Case 1: $2x - 3 = 7 \Rightarrow x =$ _____
Case 2: $2x - 3 = -7 \Rightarrow x =$ _____
Answer: _____

10. $|3x + 2| = 11$
Case 1: $3x + 2 = 11 \Rightarrow x =$ _____
Case 2: $3x + 2 = -11 \Rightarrow x =$ _____

Answer: _____

Topic 3 Solving Inequalities

Inequality Symbols: $<$ less than $>$ greater than $\leq$ less than or equal to $\geq$ greater than or equal to

Rules: Same as equations EXCEPT —

- Multiply or divide by a **positive** number: symbol stays the same.
- Multiply or divide by a **negative** number: **FLIP the symbol!**

Example: $-2x > 6 \div (-2) \Rightarrow x < -3$ (symbol flipped from $>$ to $<$)

Section 3A Solving by Adding (8 problems)

Example: Solve $x - 4 < 8$

Add 4 to both sides to undo the subtraction:

$$x - 4 + 4 < 8 + 4 \qquad \Longrightarrow \qquad \boxed{x < 12}$$

The symbol does NOT flip because we are adding (not multiplying/dividing by a negative).

1. $x - 2 < 5$

Add _____ to both sides:

x __ _____

2. $x - 4 > 1$

Add _____ to both sides:

x __ _____

3. $x - 1 \leq 6$

Add _____ to both sides:

x __ _____

4. $x - 3 \geq -2$

Add _____ to both sides:

x __ _____

5. $x - 5 < -1$

Add _____ to both sides:

x __ _____

6. $x - 7 > 0$

Add _____ to both sides:

x __ _____

x __ _____

7. $x - 2 \leq -4$

Add _____ to both sides:

8. $x - 6 \geq 3$

Add _____ to both sides:

x __ _____

Section 3B Solving by Subtracting

(8 problems)

Example: Solve $x + 5 > 9$

Subtract 5 from both sides:

$$x + 5 - 5 > 9 - 5 \qquad \Longrightarrow \qquad \boxed{x > 4}$$

Symbol stays the same (subtracting a positive).

1. $x + 3 > 7$

Subtract _____ from both sides:

x __ _____

4. $x + 4 \leq -1$

Subtract _____ from both sides:

x __ _____

2. $x + 5 < 2$

Subtract _____ from both sides:

x __ _____

5. $x + 6 > 0$

Subtract _____ from both sides:

x __ _____

3. $x + 1 \geq 8$

Subtract _____ from both sides:

x __ _____

6. $x + 2 < -3$

Subtract _____ from both sides:

x __ _____

7. $x + 8 \geq 5$

Subtract _____ from both sides:

x __ _____

8. $x + 3 \leq -6$

Subtract _____ from both sides:

x __ _____

Section 3C Solving by Multiplying (8 problems)

Example A (positive): Solve $\frac{x}{3} < 4$

Multiply both sides by 3:

$$3 \cdot \frac{x}{3} < 3 \cdot 4 \qquad \Longrightarrow \qquad \boxed{x < 12} \quad \textit{(symbol stays)}$$

Example B (negative — flip!): Solve $\frac{x}{-2} < 3$

Multiply both sides by -2 and FLIP the symbol:

$$x > -6 \quad \textit{(symbol flipped from } < \textit{ to } >\textit{)}$$

1. $\frac{x}{2} < 5$

Multiply by _____:

x __ _____ *(flip? Y/N)*

2. $\frac{x}{3} > 4$

Multiply by _____:

x __ _____ *(flip? Y/N)*

3. $\frac{x}{4} \leq -2$

Multiply by _____:

x __ _____ *(flip? Y/N)*

4. $\frac{x}{5} \geq 3$

Multiply by _____:

x __ _____ *(flip? Y/N)*

5. $\frac{x}{2} > -4$

Multiply by _____:

x __ _____ *(flip? Y/N)*

6. $\frac{x}{-2} < 3$

Multiply by -2 **(FLIP!)**:

x __ _____

7. $\frac{x}{-3} \geq 2$

Multiply by -3 **(FLIP!)**:

x __ _____

8. $\frac{x}{4} < -1$

Multiply by _____:

x __ _____ *(flip? Y/N)*

Section 3D Solving by Dividing (8 problems)

Example A (positive): Solve $3x > 12$

Divide both sides by 3:

$$\frac{3x}{3} > \frac{12}{3} \qquad \Longrightarrow \qquad \boxed{x > 4} \quad \textit{(symbol stays)}$$

Example B (negative — flip!): Solve $-4x \leq 8$

Divide both sides by -4 and FLIP the symbol:

$$x \geq -2 \quad \textit{(symbol flipped from } \leq \textit{ to } \geq\textit{)}$$

1. $2x > 8$

Divide by _____:

x __ _____ *(flip? Y/N)*

2. $3x < 15$

Divide by _____:

x __ _____ *(flip? Y/N)*

3. $4x \geq -12$

Divide by _____:

x __ _____ *(flip? Y/N)*

4. $5x \leq 20$

Divide by _____:

x __ _____ *(flip? Y/N)*

5. $-2x > 6$

Divide by -2 **(FLIP!)**:

x __ _____

6. $-3x < 9$

Divide by -3 **(FLIP!)**:

x __ _____

7. $-4x \geq 8$

Divide by -4 **(FLIP!)**:

x __ _____

8. $-5x \leq -10$

Divide by -5 **(FLIP!)**:

x __ _____

Answer Key (Teacher Copy)

Topic 1A — One-Step Equations

1. $x + 5 = 9$ — $x = 4$
2. $x - 3 = 7$ — $x = 10$
3. $2x = 14$ — $x = 7$
4. $\frac{x}{4} = 3$ — $x = 12$
5. $x + 8 = 2$ — $x = -6$
6. $x - 6 = -1$ — $x = 5$
7. $3x = -12$ — $x = -4$
8. $\frac{x}{5} = -2$ — $x = -10$
9. $-x = 9$ — $x = -9$
10. $x + 4 = -3$ — $x = -7$

Topic 1B — Two-Step Equations

1. $2x + 3 = 11$ — $x = 4$
2. $3x - 5 = 10$ — $x = 5$
3. $4x + 1 = 17$ — $x = 4$
4. $2x - 7 = 1$ — $x = 4$
5. $5x + 2 = -8$ — $x = -2$
6. $3x - 4 = -13$ — $x = -3$
7. $\frac{x}{2} + 3 = 7$ — $x = 8$
8. $\frac{x}{3} - 2 = 4$ — $x = 18$
9. $2x + 9 = 3$ — $x = -3$
10. $4x - 3 = 9$ — $x = 3$

Topic 2 — Absolute Value Equations

1. $|x| = 5$ — $x = 5$ or $x = -5$
2. $|x| = 0$ — $x = 0$
3. $|x| = -2$ — No solution
4. $|x - 3| = 4$ — $x = 7$ or $x = -1$
5. $|x + 2| = 6$ — $x = 4$ or $x = -8$
6. $|x - 1| = 0$ — $x = 1$
7. $|x + 4| + 2 = 9$ — $x = 3$ or $x = -11$
8. $2|x| = 10$ — $x = 5$ or $x = -5$
9. $|2x - 3| = 7$ — $x = 5$ or $x = -2$
10. $|3x + 2| = 11$ — $x = 3$ or $x = -\frac{13}{3}$

Topic 3A — Inequalities: Adding

1. $x - 2 < 5$ — $x < 7$
2. $x - 4 > 1$ — $x > 5$
3. $x - 1 \leq 6$ — $x \leq 7$
4. $x - 3 \geq -2$ — $x \geq 1$
5. $x - 5 < -1$ — $x < 4$
6. $x - 7 > 0$ — $x > 7$
7. $x - 2 \leq -4$ — $x \leq -2$
8. $x - 6 \geq 3$ — $x \geq 9$

Topic 3B — Inequalities: Subtracting

1. $x + 3 > 7$	$x > 4$	**5.** $x + 6 > 0$	$x > -6$
2. $x + 5 < 2$	$x < -3$	**6.** $x + 2 < -3$	$x < -5$
3. $x + 1 \geq 8$	$x \geq 7$	**7.** $x + 8 \geq 5$	$x \geq -3$
4. $x + 4 \leq -1$	$x \leq -5$	**8.** $x + 3 \leq -6$	$x \leq -9$

Topic 3C — Inequalities: Multiplying

1. $\frac{x}{2} < 5$	$x < 10$	**5.** $\frac{x}{2} > -4$	$x > -8$
2. $\frac{x}{3} > 4$	$x > 12$	**6.** $\frac{x}{-2} < 3$	$x > -6$ (*flipped*)
3. $\frac{x}{4} \leq -2$	$x \leq -8$	**7.** $\frac{x}{-3} \geq 2$	$x \leq -6$ (*flipped*)
4. $\frac{x}{5} \geq 3$	$x \geq 15$	**8.** $\frac{x}{4} < -1$	$x < -4$

Topic 3D — Inequalities: Dividing

1. $2x > 8$	$x > 4$	**5.** $-2x > 6$	$x < -3$ (*flipped*)
2. $3x < 15$	$x < 5$	**6.** $-3x < 9$	$x > -3$ (*flipped*)
3. $4x \geq -12$	$x \geq -3$	**7.** $-4x \geq 8$	$x \leq -2$ (*flipped*)
4. $5x \leq 20$	$x \leq 4$	**8.** $-5x \leq -10$	$x \geq 2$ (*flipped*)